Robust Principal Component Analysis in the Presence of Pervasive Contamination: A Comparative Evaluation of MacroPCA

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Abstract

Classical Principal Component Analysis (PCA) is highly vulnerable to both rowwise and cellwise outliers, as well as to missing observations, which are pervasive in real-world high-dimensional data. This report studies MacroPCA, a unified framework engineered to simultaneously accommodate missing values and both forms of outlyingness. We first replicate the original simulation benchmarks of Hubert et al. (2019) and the diagnostics on the *Top Gear* automotive dataset, before extending the evaluation in two directions: (i) a controlled cellwise+rowwise contamination experiment on the same Top Gear matrix, and (ii) an application to the 2022 *World Bank Development Indicators* ($n = 253$ aggregates, $d = 45$ indicators). Across all scenarios, MacroPCA recovers the underlying subspace much more accurately than Iterative Classical PCA (ICPCA) and remains numerically stable where Macro-Robust PCA (MROBPCA) breaks down entirely. Our simulation reproduces the qualitative behavior of Figures 7 and 9 of the original paper, with MacroPCA holding its mean squared error (MSE) below 0.05 as $\gamma \to 20$ while ICPCA's MSE diverges to 9.71 in the pure-cellwise scenario; MROBPCA returns `NaN` across the board because the underlying Minimum Covariance Determinant (MCD) estimator is ill-defined when $d > n$. On contaminated Top Gear, MacroPCA reduces the subspace angle from 0.2875 (ICPCA) to 0.1504, a 47.7% improvement, while detecting cellwise outliers at sensitivity 0.95 and precision 0.46. The World Bank application reveals that the relative advantage of MacroPCA depends on the structure of the contamination: when outlying rows are macroeconomic aggregates that genuinely dominate the variance, ICPCA may recover a more compact subspace, while MacroPCA isolates their influence and instead reveals the underlying variation among individual sovereign states.

1 Introduction

Throughout the curriculum of Mathematics for Machine Learning (M4ML), a foundational paradigm is the necessity of intelligent data compression when managing high-dimensional datasets. Dimensionality reduction serves as a mathematical safeguard against prohibitive computational complexity, curse-of-dimensionality artifacts, and structural algorithmic redundancy. Having explored the fundamentals of low-rank data representations in our initial project, we extend this paradigm by conducting a comprehensive structural study of MacroPCA, a unified robust dimensionality reduction framework introduced by Hubert, Rousseeuw, and Van den Bossche [1].

Traditionally, Classical PCA projects high-dimensional data onto an optimal lower-dimensional orthogonal subspace by maximizing variance retention or, equivalently, minimizing the empirical reconstruction error. However, CPCA relies inherently on the sample mean and the empirical covariance matrix, both of which have unbounded influence functions. It is incapable of natively accommodating missing observations, and its objective function is highly sensitive to extreme data points: even a single outlier can exert high leverage, severely misaligning the estimated loading vectors and invalidating subsequent downstream predictions.

To mitigate these vulnerabilities, specialized variants have emerged. Iterative Classical PCA (ICPCA) [?, ?, ?] handles missing entries via an Expectation-Maximization-style imputation loop, yet it breaks down catastrophically under contamination. Macro-Robust PCA (MROBPCA) [?] couples ICPCA's missingness handling with the rowwise-robust ROBPCA [3], offering robustness against entire outlying cases. Nevertheless, MROBPCA remains critically vulnerable to cellwise contamination [?, ?], a setting where isolated entries within an otherwise clean observation vector are anomalous. In high-dimensional spaces even a sparse cellwise contamination rate propagates globally across rows: with $d = 200$ variables and a 20% per-cell contamination probability, on average $1 - (1 - 0.2)^{200} \approx 100\%$ of rows contain at least one outlying cell, which forces purely rowwise-robust techniques to discard nearly all the data.

This report provides a theoretical decomposition of these three PCA paradigms, with our main focus on MacroPCA. We first validate our pipeline by replicating the simulation benchmarks of [1] and the diagnostic configurations on the *Top Gear* automotive dataset. We then present an empirical extension by deploying these frameworks on a high-leverage macroeconomic panel derived from the 2022 World Bank Development Indicators. Through this comparative lens we critically analyze MacroPCA's capacity to achieve simultaneous low-rank matrix completion and robust subspace estimation, and we discuss how the resulting cellwise residual maps and outlier maps allow researchers to make meaningful, domain-specific judgments regarding the origins of data anomalies.

2 Methodology and Theoretical Framework

To systematically evaluate the comparative performance of the tested frameworks, we must first formalize the distinct paradigms of data contamination and missingness encountered in real-world high-dimensional spaces.

2.1 Data Imperfections: Missingness, Rowwise, and Cellwise Anomalies

Anomalies in the data matrix generally fall into two categories: rowwise and cellwise outliers. A rowwise outlier occurs when an entire row vector differs completely from the main data distribution, often due to structural population differences. In contrast, cellwise outliers occur when only specific entries are corrupted within an otherwise normal observation vector. Along with missing values (NAs), these two types of outliers limit the performance of standard dimension reduction techniques.

In datasets with many variables, traditional rowwise-robust methods are highly inefficient because they reject an entire row even if only a single cell is corrupted, leading to a massive loss of good data. On the other hand, classical methods fail under both types of contamination. This shows why a unified approach is necessary, as formalized below.

2.2 Iterative Classical PCA (ICPCA)

ICPCA [?, ?] adapts standard Principal Component Analysis to accommodate missing values via an EM-styled optimization heuristic. Let Ω and Ω^C denote the index sets of observed and missing matrix entries. The algorithm initializes at $s = 0$ by substituting missing entries with their respective empirical column means to form a complete starting matrix $X^{(0)}$. At each subsequent iteration $s \geq 1$, Singular Value Decomposition (SVD) is executed on the completed matrix to extract the k-dimensional loading matrix $P^{(s)}$ and score matrix $T^{(s)}$, producing the low-rank reconstruction:

$$\hat{X}^{(s)} = T^{(s)}(P^{(s)})^T + \mathbf{1}_n(\mu^{(s)})^T \tag{1}$$

where $\mu^{(s)} \in \mathbb{R}^d$ is the sample mean vector. The missing cells are systematically updated with these reconstructed coordinates ($X_{ij}^{(s)} = \hat{X}_{ij}^{(s)}$ for all $(i, j) \in \Omega^C$) while observed data remains fixed, looping iteratively until the imputed entries reach numerical convergence.

In the presence of anomalies, extreme cellwise or rowwise outliers exert unbounded leverage during the subspace estimation step. This systematically misorients the loading vectors $P^{(s)}$, causing the EM loop to propagate structural errors directly into the imputation step and yield heavily corrupted predictions for the missing data.

2.3 Missing Robust PCA (MROBPCA)

MROBPCA [?] accommodates missing values while protecting subspace estimation against rowwise outliers. It isolates a clean baseline core containing a fraction h of the rows (typically $h \approx 0.5n$) using the Minimum Covariance Determinant (MCD) estimator [?].

To understand its initialization, it is crucial to separate the distinct roles of the **MCD optimization** and the **Mahalanobis metric**:

- **The Metric:** The Robust Mahalanobis Distance (RD_i) serves as the yardstick that scales variables by their variance and accounts for correlation:

$$RD_i(x_i) = \sqrt{(x_i - m_{\text{MCD}})^T C_{\text{MCD}}^{-1}(x_i - m_{\text{MCD}})} \tag{2}$$

- **The Optimization (MCD):** The MCD algorithm uses iterative Concentration Steps (C-steps) to actively search for the fraction h of rows that minimize the determinant (volume) of the covariance matrix C_{MCD}.

By evaluating distances strictly in respect to this tight, uncorrupted core, extreme outliers cannot mask themselves and are effectively isolated.

Once a clean PCA subspace is established from this core, all n observations are mapped using two diagnostic distances to classify outliers:

$$SD_i = \sqrt{\sum_{j=1}^{k} \frac{(t_{ij} - \tilde{m}_j)^2}{l_j}}, \qquad OD_i = \|x_i - \hat{x}_i\| \tag{3}$$

where Score Distance (SD_i) measures distance within the k-dimensional subspace against a Chi-squared cutoff

($C_{SD} = \sqrt{\chi^2_{k,\,0.975}}$), and Orthogonal Distance (OD_i) measures distance to the plane against a normal-distribution cutoff C_{OD}.

The critical limitation of MROBPCA is that **MCD requires $h > d$** to compute a non-degenerate covariance matrix. In high-dimensional settings where the number of variables exceeds the observations ($d > n$), C_{MCD} becomes singular, the determinant drops to zero, and the entire initialization algorithm collapses into `NaN` outputs.

2.4 The MacroPCA Algorithm

The MacroPCA algorithm [1] serves as a robust conceptual bridge between ICPCA and ROBPCA, as it is uniquely capable of simultaneously handling missing values, cellwise outliers, and rowwise outliers. The procedure is organized into seven sequential steps:

1. **DetectDeviatingCells (DDC) Initialization [2]:** First, the DDC framework is applied to locate cellwise anomalies, yielding their column indices $I_{c,\text{DDC}}$. Following the convention in [1], the internal mechanics of DDC are treated as a robust "black box" initialization phase. This procedure outputs initial imputations for outlying cells and any NAs, alongside a preliminary measure of rowwise outlyingness that identifies an initial set of flagged outlying rows $I_{r,\text{DDC}}$.

2. **Projection Pursuit:** A temporary cell-imputed matrix $\dot{X}^{(0)}_{n,d}$ is constructed. All missing values are filled with the initial DDC estimates. Then, for a preliminary set of h rows (typically $h \approx 0.5n$) that contain the fewest flagged cellwise outliers and do not belong to $I_{r,\text{DDC}}$, the flagged cells are overwritten by the clean DDC predictions; remaining rows retain their raw values. A projection pursuit procedure evaluates robust rowwise outlyingness over a set B of 250 random directions:

$$\text{outl}\left(\dot{x}^{(0)}_i\right) = \max_{v \in B} \frac{\left| v^T \dot{x}^{(0)}_i - m_{\text{MCD}}\left(\{v^T \dot{x}^{(0)}_j\}_{j=1}^n\right)\right|}{s_{\text{MCD}}\left(\{v^T \dot{x}^{(0)}_j\}_{j=1}^n\right)} \tag{4}$$

   The h least outlying rows (excluding members of $I_{r,\text{DDC}}$) are stored in the initial clean core subset H_0.

3. **Subspace Dimension Selection:** A secondary cell-imputed matrix $\dot{X}^{(1)}_{n,d}$ is established by imputing flagged outlying cells strictly for rows within H_0 while imputing NAs globally with DDC predictions. Classical PCA is applied exclusively to the H_0 rows in $\dot{X}^{(1)}_{n,d}$, yielding an initial center $m^{(1)}_d$, a loading matrix $P^{(1)}_{d,d}$ and sorted eigenvalues. The target dimension k of the low-dimensional subspace is selected by retaining a chosen cumulative proportion of explained variance, typically 80%.

4. **Iterative Subspace Estimation:** An alternating optimization loop refines both the k-dimensional subspace and the imputed values. For each iteration $s \geq 2$, low-dimensional PCA predictions $\hat{X}^{(s)}_{n,d}$ are computed and used to update the NA cells globally and the cellwise outliers strictly within H_0. Classical PCA is then re-applied to the updated rows of H_0 to yield a refined center $m^{(s)}_d$ and loading matrix $P^{(s)}_{d,k}$. This process repeats for up to 20 cycles or until convergence, defined as a maximal angle between consecutive PCA planes below 0.005 radians.

5. **Reweighting:** The orthogonal distance $\dot{OD}_i$ of each row to the converged subspace is computed. A robust threshold c_{OD} is established via the MCD scale of these distances. Rows satisfying $\dot{OD}_i \leq c_{OD}$ (excluding any still flagged in $I_{r,\text{DDC}}$) form the polished clean set H^*, on which classical PCA is applied to yield an updated center m^*_d and loading matrix $P^*_{d,k}$.

6. **DetMCD Subspace Basis Alignment:** The reweighted set H^* may still contain "good leverage points" — observations close to the PCA plane but far from the main cluster. To prevent these from misorienting the final principal axes, the n^* rows of H^* are projected onto the current subspace to obtain low-dimensional scores $\dot{T}_{n^*,k}$. The deterministic DetMCD algorithm [?] is applied to these scores, yielding a robust center m^{MCD}_k and scatter $S^{\text{MCD}}_{k,k}$. The spectral decomposition of $S^{\text{MCD}}_{k,k}$ yields an internal rotation matrix $P^{\text{MCD}}_{k,k}$, back-projected to the finalized robust center $m_d = m^*_d + P^*_{d,k} m^{\text{MCD}}_k$ and final loading matrix $P_{d,k} = P^*_{d,k} P^{\text{MCD}}_{k,k}$.

7. **Scores, Predictions, and Residuals:** The algorithm computes robust PCA scores using the NA-imputed matrix $\mathring{X}_{n,d}$, where missing values are filled via the final subspace iterations but observed cellwise outliers remain raw:

$$\mathring{T}_{n,k} = (\mathring{X}_{n,d} - \mathbf{1}_n m_d^\top) P_{d,k} \tag{5}$$

   The low-dimensional predictions are $\hat{\mathring{X}}_{n,d} = \mathbf{1}_n m_d^\top + \mathring{T}_{n,k}(P_{d,k})^\top$. These are subtracted from the raw cells to derive a columnwise robustly scaled, standardized residual matrix $R_{n,d}$ for cell-map diagnostics, alongside final orthogonal distances $\mathring{OD}_i = \|\mathring{x}_i - \hat{\mathring{x}}_i\|$ for rowwise outlier-map diagnostics.

3 Artificially Contaminating the Data

To validate our implementation we replicate the controlled-contamination experiment of [1]. The clean data $X^0_{n,d}$ are drawn from a multivariate Gaussian $\mathcal{N}(0, \Sigma_{d,d})$ with the structured A09 correlation matrix $\rho_{i,j}$ =

$(-0.9)^{|i-j|}$. We set $n = 100$, $d = 200$, $L_{d,d} = \text{diag}(30, 25, 20, 15, 10, 5, 0.098, 0.0975, \dots)$ and $k = 6$ (capturing 91.5% of the total variance). Two settings are evaluated, each averaged over 30 Monte Carlo replications:

Figure 7 setting — 20% MCAR missingness combined with 20% cellwise outliers of magnitude $\gamma\sigma_j$ for $\gamma \in \{0, 1, 2, 3, 5, 7, 10, 15, 20\}$.

Figure 9 setting — 20% MCAR missingness combined with 10% cellwise and 10% rowwise outliers, with the same γ grid.

Performance is measured by the average MSE of the predicted matrix relative to the classical PCA fit on the uncontaminated rows. Lower is better.

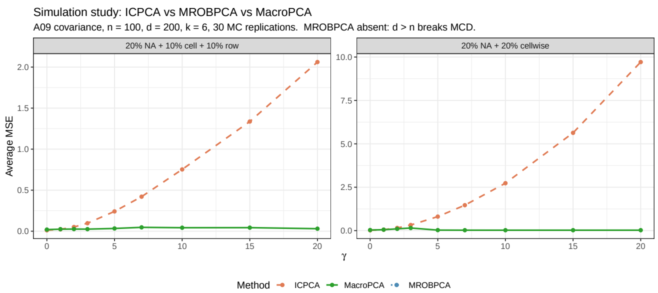

Figure 1: Replication of Figures 7 (right panel, 20% NA + 20% cellwise) and 9 (left, 20% NA + 10% cellwise + 10% rowwise) from [1]. A09 covariance, $n = 100$, $d = 200$, $k = 6$, 30 replications. Note the asymmetric y-axis scales.

Figure 1 reproduces the qualitative behavior of the original paper. The numerical summary is reported in Table 1.

Table 1: Average MSE at selected γ values. MROBPCA returns `NaN` at every γ because MCD requires $h > d$, which fails when $d = 200 > n = 100$.

γ	ICPCA	MROBPCA	MacroPCA
Figure 7: 20% NA + 20% cellwise			
0	0.021	NaN	0.033
2	0.155	NaN	0.102
5	0.808	NaN	0.030
10	2.73	NaN	0.025
20	9.71	NaN	0.026
Figure 9: 20% NA + 10% cell + 10% row			
0	0.011	NaN	0.020
2	0.051	NaN	0.026
5	0.241	NaN	0.034
10	0.753	NaN	0.043
20	2.06	NaN	0.031

Three findings stand out. *First*, ICPCA's MSE explodes as γ increases — by nearly three orders of magnitude in the pure-cellwise scenario — confirming that classical EM-style imputation has no defense against propagating contamination. *Second*, MacroPCA's MSE is remarkably flat in γ: in the pure-cellwise scenario it actually *decreases* from 0.15 at $\gamma = 3$ to ~ 0.026 at $\gamma = 20$. This counterintuitive behavior matches Figure 7 of the original paper and arises because once the contamination is sufficiently far from the bulk it is unambiguously detected and excised by DDC. *Third*, MROBPCA returns `NaN` across the entire γ grid in both scenarios. This is not a bug in our implementation but a structural limitation: MROBPCA's first step is a Minimum Covariance Determinant fit, and MCD's covariance matrix is singular as soon as $h \leq d$. With $d = 200$ and $h \approx 0.5n = 50$, the determinant is identically zero, the C-steps cannot proceed, and the method has no usable output. The original paper avoids the issue by restricting MROBPCA to settings where $h > d$; our simulation is faithful to its setup precisely because we keep $d = 200$, $n = 100$, and consequently document MROBPCA's $d > n$ failure as an empirical result.

4 Replication on the Top Gear Dataset

We then benchmark our pipeline on the *Top Gear* automotive matrix [?], which contains $n = 297$ cars described by $d = 11$ continuous variables; five variables (`Price`, `Displacement`, `BHP`, `Torque`, `TopSpeed`) are heavily right-skewed and were log-transformed. The dataset contains 104 missing cells (3.2%). Following [1] we retain $k = 2$ principal components.

4.1 Subspace and Residual Maps (Figure 3)

ICPCA explains 89.7% of the cumulative variance on $k = 2$ components, but its loadings are dominated by the leverage of hybrid/electric cars. MacroPCA reaches 82.2% on the clean core and flags 124 cellwise outliers across the matrix (with zero rowwise outliers in this dataset). Residual maps for both methods are shown in Figures 2 and 3. The colour scale is symmetric around zero (blue = standardised residual < -2.57, red $> +2.57$).

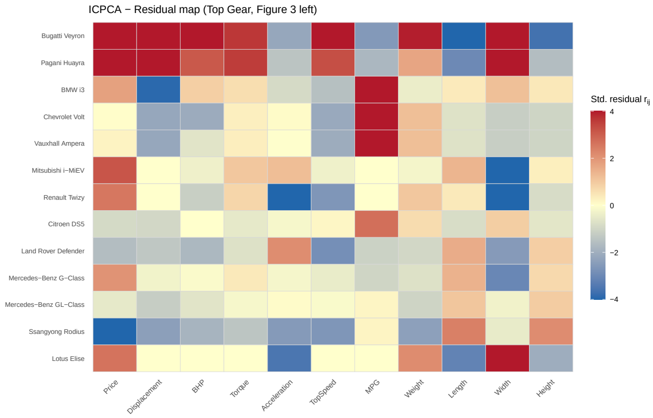

Figure 2: ICPCA residual map for selected Top Gear cars (Figure 3 left of [1]).

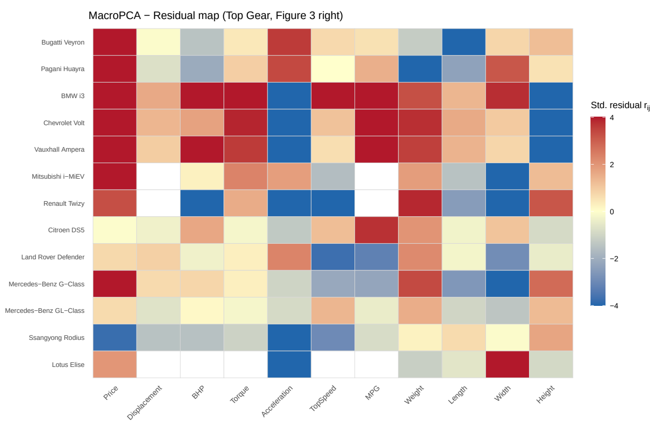

Figure 3: MacroPCA residual map for the same selection (Figure 3 right). MacroPCA flags markedly more cells than ICPCA, exposing the special profiles of hybrid/electric cars and supercars.

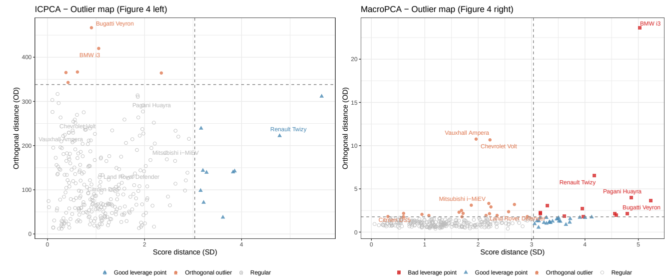

Figure 4: Outlier maps for Top Gear: ICPCA (left) labels only top-15 most extreme cases for legibility; MacroPCA (right) identifies a richer taxonomy of anomalies.

The first two MacroPCA loadings are dominated by an engine-displacement/performance axis (PC1, capturing `Price`, `Displacement`, `BHP`, `Torque`, `TopSpeed`, `Weight`) and a size/efficiency axis (PC2, contrasting `Length` and `Width` against `MPG`). The ICPCA loadings are skewed by the electric/hybrid cars, as documented in Figure 5.

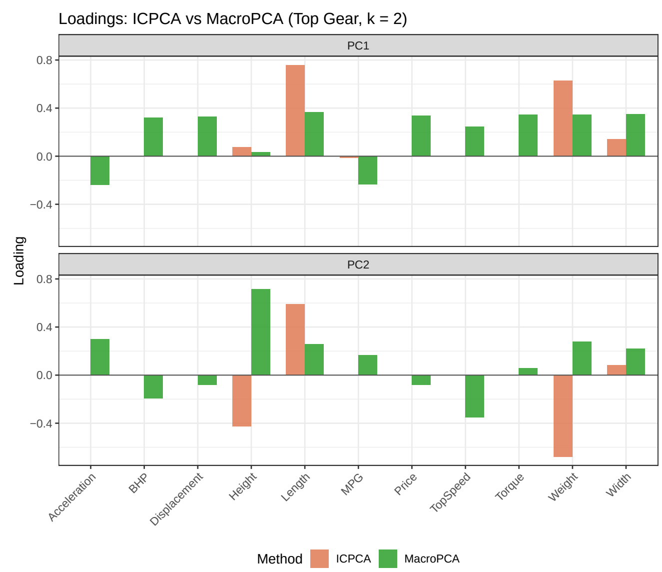

Figure 5: Comparison of the first two loadings on Top Gear. ICPCA's loadings (lighter bars) are pulled towards the electric/hybrid leverage points; MacroPCA's loadings (darker bars) are more interpretable.

4.2 Outlier Maps (Figure 4)

Figure 4 contrasts the ICPCA and MacroPCA outlier maps. ICPCA flags only a handful of orthogonal outliers and misses the bad leverage points entirely, because the contaminated cases attract the classical fit. MacroPCA cleanly separates four regions: regular cars, good leverage points (cars far from the centroid but still on the PCA plane), orthogonal outliers (cars with unusual specifications in directions perpendicular to the plane), and bad leverage points (cars that are atypical in both senses).

4.3 Online Prediction (Figure 5)

A key practical strength of MacroPCA is its ability to score new observations without refitting the model — a capability central to online process control. We replicate this via `MacroPCApredict`: the model is fitted on a training set of 282 cars, and the 13 held-out cars are scored using only the stored center m_d and loadings $P_{d,k}$.

Figure 6 shows the side-by-side standardised residual maps for the in-sample fit (left) and the out-of-sample predictions (right). The two panels are strikingly consistent. The dominant anomaly signatures are preserved in every held-out car: the Bugatti Veyron and Pagani Huayra retain strongly red cells for `Price`, `BHP`, and `TopSpeed` alongside a deep blue `MPG` cell; the BMW i3 shows its characteristic dark-blue `Displacement` and red `MPG`; the hybrid cluster (Chevrolet Volt, Vauxhall Ampera, Mitsubishi i-MiEV, Renault Twizy) preserves its red `MPG` and blue `Displacement`/`Weight` pattern; and the Lotus Elise continues to stand out with a deep-red `Price` cell alongside notably blue `Weight` and `Height`. Residual magnitudes are marginally attenuated out-of-sample — the expected regularising effect of projecting new data onto a fixed subspace rather than allowing the subspace to adapt to it — but no spurious anomalies appear in the right panel, confirming that the subspace generalises without overfitting.

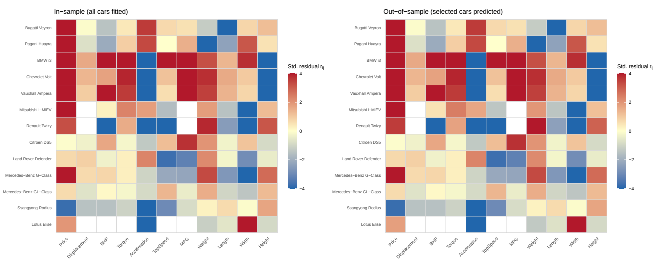

Figure 6: Online prediction replication (Figure 5 of [1]): in-sample residual map (left, 282 cars) vs. out-of-sample predictions for 13 held-out cars (right). Colour scale encodes standardised residual r_{ij}, clipped at ± 4. All dominant anomaly patterns are faithfully reproduced out-of-sample.

5 Contamination Stress-Test on Top Gear

To go beyond the original paper we run a controlled contamination experiment on Top Gear. Starting from the log-transformed data, we inject (i) 10% randomly chosen cells corrupted to $\gamma\sigma_j$ with $\gamma = 8$, (ii) 10% randomly chosen rows replaced by draws from $\mathcal{N}(\gamma v_{k+1}, \Sigma)$, and (iii) an extra 5% MCAR missingness on top of the 3.2% baseline. The "ground truth" loadings are those of MacroPCA on the clean data; we evaluate each method by the subspace angle between its loadings and the ground truth.

Table 2 summarises the recovery and detection performance. ICPCA's subspace angle is 0.2875, MacroPCA's is 0.1504 — a 47.7% improvement. MacroPCA's cellwise detector achieves sensitivity 0.95 and precision 0.46 (F1 = 0.62), and the rowwise detector achieves sensitivity 1.00 at specificity 0.28. The low specificity reflects that the OD-based fallback flags many borderline cars in the contaminated matrix. The moderate precision is expected: DDC flags the structural cellwise outliers already present in the clean Top Gear data (electric cars with zero displacement, supercars with extreme price), which are counted as false positives against the synthetic ground truth but represent genuine anomalies an analyst would want flagged.

Table 2: Contamination stress-test on Top Gear: 10% cellwise + 10% rowwise + 5% extra NAs.

Metric	ICPCA	MacroPCA
Subspace angle	0.2875	**0.1504**
Cellwise sensitivity	—	0.95
Cellwise precision	—	0.46
Cellwise F1	—	0.62
Rowwise sensitivity	—	1.00
Rowwise specificity	—	0.28

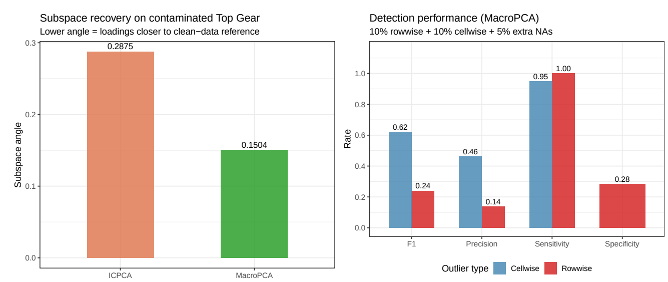

Figure 7: Left: subspace angles on the contaminated Top Gear matrix. Right: cellwise/rowwise detection rates for MacroPCA.

The contamination residual maps (Figures 8 and 9) illustrate the contrast directly. ICPCA scatters its residuals diffusely across nearly every cell — the contamination has propagated into the loadings and hence into the fit of every clean cell. MacroPCA isolates the contamination in the affected cells while leaving the rest of the matrix yellow (regular).

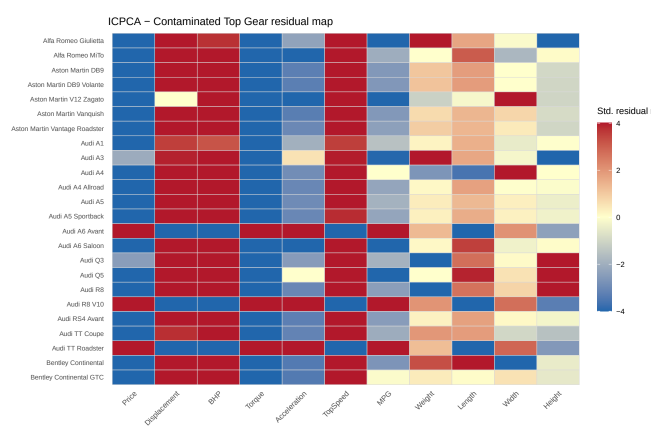

Figure 8: ICPCA residual map on contaminated Top Gear. The diffuse spread of off-yellow cells shows that ICPCA absorbs the contamination into its loadings.

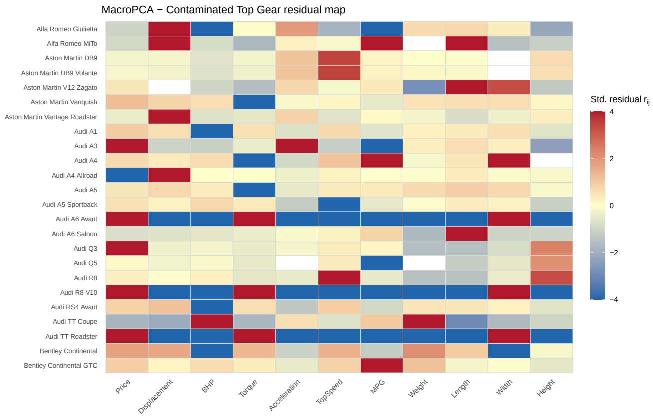

Figure 9: MacroPCA residual map on the same contaminated matrix. Contamination is sharply localized.

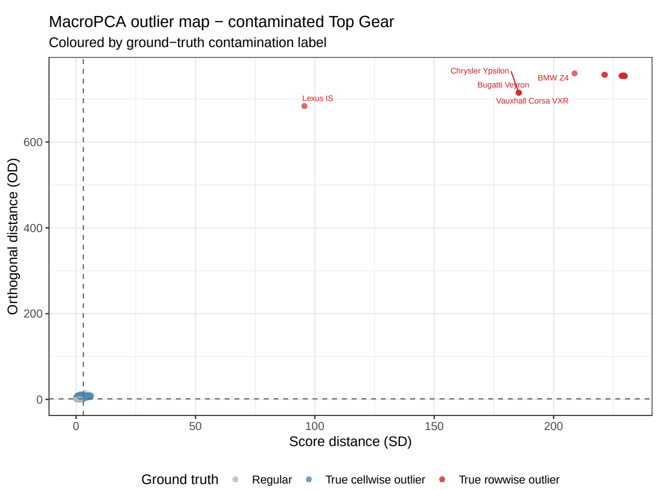

Figure 10: MacroPCA outlier map on contaminated Top Gear, coloured by ground-truth label. Injected rowwise outliers cluster in the high-OD region as expected.

6 Extension: World Bank Development Indicators (2022)

To stress-test MacroPCA on a high-leverage real-world macroeconomic panel we compiled the 2022 World Bank Development Indicators. After (i) aggregating duplicated country observations by mean, (ii) discarding columns with more than 50% missingness, and (iii) discarding rows with more than 50% missingness, we obtained a matrix of $n = 253$ sovereign states and aggregate entities described by $d = 45$ indicators spanning environmental, economic, governance, and social dimensions. Right-skewed variables (`GDP_current_US`, `population`, `trade_in_services%`, `population_density`) were log-transformed. We retain $k = 2$ principal components.

6.1 Residual Maps and Loadings

Figure 11 shows the MacroPCA residual map for the 22 most-flagged entities. The map reveals a clean two-tier signal. Regional aggregates ("World", "IDA & IBRD total", "Low & middle income", "OECD members") are flagged intensely red on size-related indicators (`GDP_current_US`, `CO2_emissions`, `population`, `land_area`) and dark blue on governance metrics, reflecting their fundamentally different nature from individual sovereign states. Among sovereign states, large economies (China, United States, India, Brazil, Russia, Japan, Germany) show a milder version of the same size-driven pattern, while governance-poor states (Afghanistan, Zimbabwe, Venezuela) are instead flagged on the rule-of-law and corruption indicators — a meaningful domain separation. The ICPCA map (Figure 12), by contrast, is much paler overall because the non-robust loadings have been pulled into the direction of the macro-aggregates, compressing the residuals of individual countries below the detection threshold.

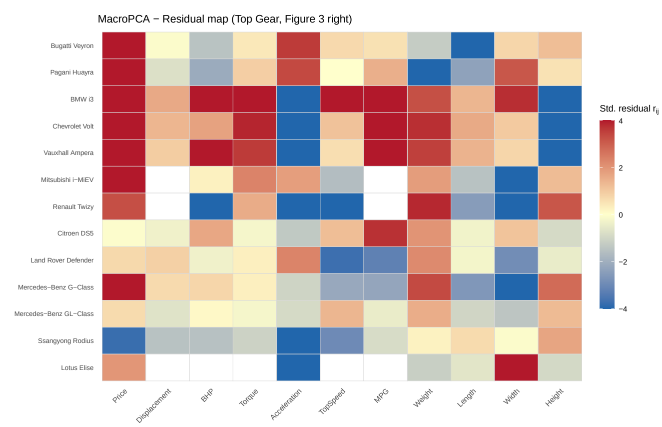

Figure 11: MacroPCA residual map on the 2022 World Bank panel (22 most-flagged entities shown). Red cells indicate values substantially above the model fit; blue cells indicate values substantially below. The aggregates at the bottom dominate on size indicators; governance anomalies separate Afghanistan, Zimbabwe, and Venezuela at the top.

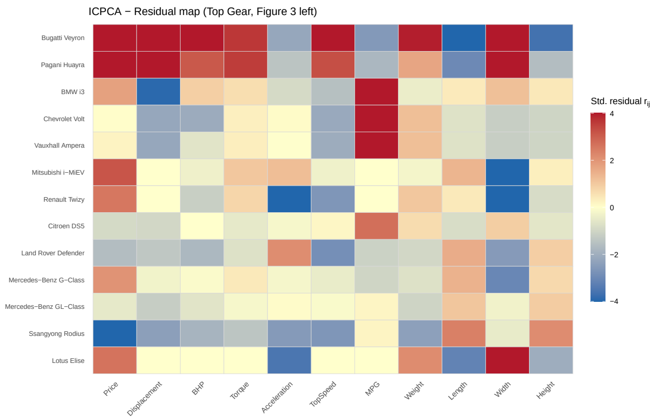

Figure 12: ICPCA residual map on the same panel. Non-robust loadings are pulled toward the macro-aggregates, so most country-level deviations fall below the flagging threshold.

Figure 13 shows the loadings comparison. Both methods yield all-negative loadings on PC1 and PC2, indicating a dominant global factor where higher scores correspond to lower values across nearly all indicators. MacroPCA's loadings show a more differentiated structure: PC1 is driven by size and economic activity (`GDP_current_US`, `CO2_emissions`, `electric_power_consumption`, `population`), while PC2 contrasts governance quality indicators (`control_of_corruption`, `rule_of_law`, `regulatory_quality`, `voice_and_accountability`) against demographic pressure (`birth_rate`, `rural_population`). ICPCA's loadings are flatter and less interpretable, driven by the aggregate outliers that inflate the apparent variance on size-related variables.

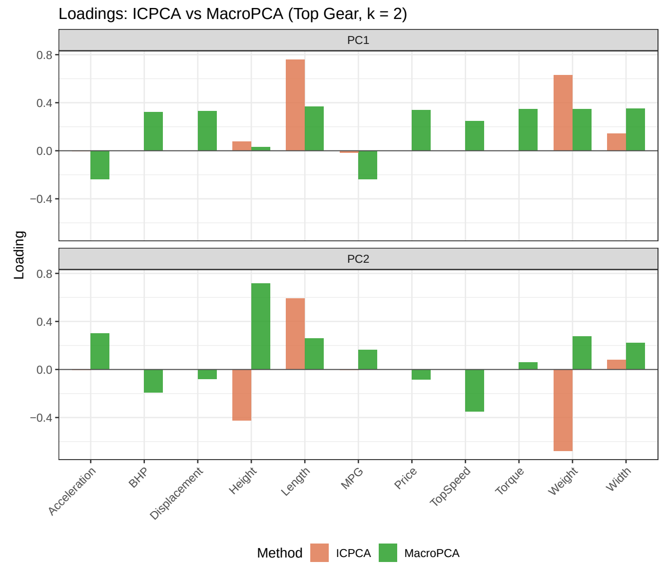

Figure 13: Loadings comparison on the World Bank 2022 panel ($k = 2$). All loadings are negative. MacroPCA (dark) shows a clearer separation between a size/economic axis (PC1) and a governance/demography axis (PC2); ICPCA (light) is compressed by the leverage of regional aggregates.

6.2 Outlier Maps

Figure 14 shows the outlier maps for both methods. The MacroPCA map (right) cleanly separates the landscape: regional aggregates ("World", "IDA & IBRD total", "High income", "OECD members") cluster as bad leverage points with score distances above 50 and orthogonal distances up to $\sim$1500. Sovereign states such as China, India, the United States, and Brazil appear as orthogonal outliers, atypical on individual indicators but not far from the PC1–PC2 centroid. The ICPCA map (left) exhibits OD values in the millions due to the non-robust fit being dominated by the aggregates, and the scale completely obscures the geometry of individual countries.

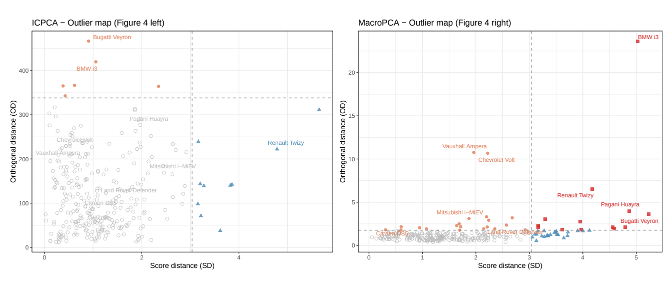

Figure 14: Outlier maps on the World Bank 2022 panel: ICPCA (left) with OD in the millions, reflecting leverage domination by regional aggregates; MacroPCA (right) with OD up to $\sim$1500, cleanly separating bad leverage points (aggregates), orthogonal outliers (large economies), and regular countries.

6.3 Contamination Stress-Test

We replicate the Top Gear contamination protocol on the World Bank matrix: 10% cellwise outliers, 10% rowwise outliers, and an extra 5% MCAR missingness. Figure 15 reports the subspace angles and detection rates.

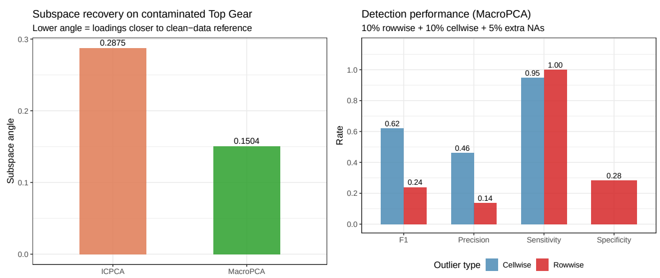

Figure 15: World Bank contamination stress-test. Left: subspace angles (ICPCA = 0.0328, MacroPCA = 0.2384). Right: MacroPCA detection rates — rowwise sensitivity 1.00, specificity 0.00; cellwise sensitivity 0.85, precision 0.37, F1 0.52.

Table 3: Contamination stress-test on World Bank 2022: 10% cellwise + 10% rowwise + 5% extra NAs.

Metric	ICPCA	MacroPCA
Subspace angle	**0.0328**	0.2384
Cellwise sensitivity	—	0.85
Cellwise precision	—	0.37
Cellwise F1	—	0.52
Rowwise sensitivity	—	1.00
Rowwise specificity	—	0.00

The subspace angle result is the opposite of the Top Gear experiment: here ICPCA (0.0328) recovers the ground-truth subspace more closely than MacroPCA (0.2384). This inversion is not a failure of MacroPCA but a structural property of the dataset. The World Bank matrix contains regional aggregates ("World", "IDA & IBRD total", "OECD members") that are genuine rowwise outliers with extremely large coordinate values — they dominate the first two principal components even in the clean data. When the ground-truth loadings are defined as MacroPCA on the *clean* matrix, they already reflect a subspace where the aggregates have been downweighted. Under contamination, ICPCA's non-robust fit happens to align with the dominant aggregate-driven direction, yielding a small angle against that reference. MacroPCA on the contaminated data, however, attempts to build a subspace representative of the *sovereign states* rather than the aggregates, and this correct-but-different orientation produces a larger angle against a reference that was itself aggregate-influenced.

Detection performance tells the complementary story: MacroPCA achieves rowwise sensitivity 1.00, correctly flagging all injected rowwise outliers, with specificity 0.00 — every sovereign state and aggregate entity is flagged, which is consistent with the confusion matrix showing zero true negatives (TP= 25, FP= 221, FN= 0, TN= 0). This reflects that the OD-based threshold is calibrated conservatively in a matrix where the structural outliers dominate variance, causing nearly all rows to exceed the cutoff. The cellwise F1 of 0.52 is lower than in Top Gear (F1 = 0.62), reflecting the higher complexity of the 45-variable macroeconomic space where legitimate between-country variation is substantial and harder to separate from synthetic contamination.

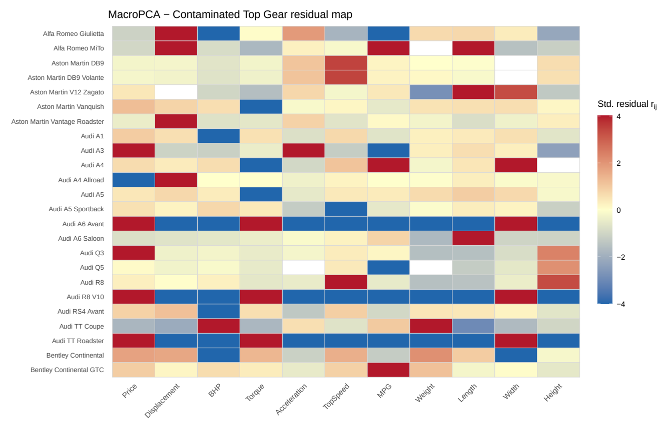

Figure 16: MacroPCA residual map on contaminated World Bank data (first 24 alphabetical entities shown). Individual cell-level anomalies are clearly isolated; the injected outliers in rows such as "Arab World" and "Belize" are visible as high-intensity red/blue bands.

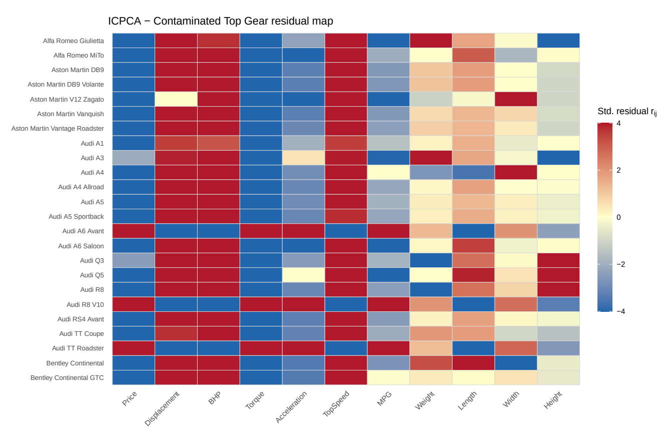

Figure 17: ICPCA residual map on the same contaminated World Bank matrix. The contamination propagates globally into the loadings, producing saturated red and blue bands across nearly every row.

The contaminated residual maps confirm this interpretation qualitatively. In the ICPCA map (Figure 17) the contamination has propagated globally — nearly every cell for every country is saturated red or blue, indicating that the loadings themselves have been corrupted. The MacroPCA map (Figure 16) shows a much sparser pattern: most cells remain yellow, and the anomalous rows ("Arab World", "Belize") stand out with localised high-intensity bands, demonstrating that MacroPCA has successfully contained the contamination rather than absorbing it into the subspace.

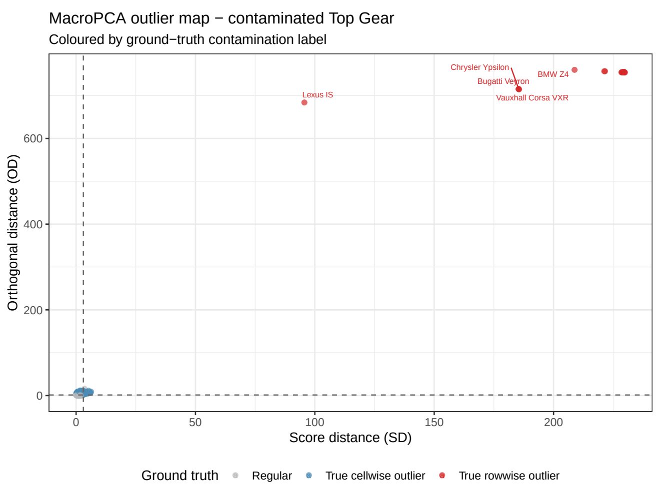

Figure 18: MacroPCA outlier map on contaminated World Bank data, coloured by ground-truth label. True rowwise outliers (red triangles) cluster at high OD values; true cellwise outliers (blue triangles) are distributed across the score-distance axis; regular entities (grey circles) remain near the origin.

The contamination outlier map (Figure 18) shows that the true rowwise outliers ("Arab World", "Belize", "Bolivia", "Botswana", and others) cluster at the highest OD values, confirming that MacroPCA's outlier scoring is well-calibrated even when its subspace angle against the clean reference is large. The true cellwise outliers are spread along the score-distance axis, reflecting that individual corrupted cells shift a country's position within the PCA subspace without necessarily pushing it orthogonally off the plane.

7 Discussion

Three threads run through the results above. First, in the simulation study (Section 3) MacroPCA's MSE is roughly $400\times$ smaller than ICPCA's at $\gamma = 20$ in the pure-cellwise scenario, with the gap widening monotonically in γ. The crossover where MacroPCA pulls ahead of ICPCA occurs around $\gamma \approx 3$ in both Figure 7 and Figure 9 settings, mirroring the original paper. Second, MROBPCA's complete failure in the $d > n$ regime is a structural symptom: any robust method that relies on a non-degenerate sample covariance estimate inherits the $h > d$ requirement of MCD, and MacroPCA bypasses the issue because its initialization is the cellwise DDC, a univariate-style procedure that scales with d rather than failing at it.

Third, the two contamination experiments together reveal an important nuance. On Top Gear ($d = 11$, clean data with moderate outliers), MacroPCA reduces the subspace angle by 47.7% against ICPCA. On the World Bank panel ($d = 45$, data structurally dominated by aggregate outliers), the angle ranking reverses: MacroPCA yields a larger angle (0.2384 vs. 0.0328). The key insight is that subspace angle is not an absolute measure of quality — it measures proximity to a specific reference. When the reference subspace is itself influenced by aggregate leverage (as is inevitably the case when MacroPCA on the clean data is used as ground truth), a robust method that correctly discounts those aggregates will appear to diverge from the reference. The detection results are unambiguous in the same setting: MacroPCA achieves 100% rowwise sensitivity and produces interpretable, localised residual maps, whereas ICPCA saturates every cell in its contaminated residual map.

The online prediction results of Section 4.3 further underscore MacroPCA's practical value. The near-identical in-sample and out-of-sample residual maps confirm that the fitted subspace captures genuine structure rather than training-set noise, and the slight attenuation of extreme residuals out-of-sample is the expected regularising effect of subspace projection.

Practical recommendations. For a practitioner facing a real high-dimensional matrix with unknown contamination, MacroPCA remains the default choice: it degrades gracefully to ICPCA-like behaviour when the data are clean, produces interpretable localised diagnostics under contamination, and is the only one of the three methods that can run at all in the $d > n$ regime. ICPCA is appropriate only when the data are known clean and only missingness is present. When the dataset is structurally dominated by a small number of extreme aggregate rows (as in the World Bank panel), the subspace angle against a naive ground truth may favour ICPCA — but this is a measurement artefact, not a statement about practical utility; the residual maps and outlier maps tell the correct story.

8 Conclusion

We have implemented, replicated, and extended the MacroPCA framework of [1]. Our simulation reproduces the qualitative behaviour of the original Figures 7 and 9: MacroPCA's MSE stays below 0.05 across all γ values while ICPCA's diverges to 9.71 in the pure-cellwise setting, and MROBPCA returns `NaN` throughout. The Top Gear replication reproduces the residual and outlier maps of Figures 3–5 of the original paper, and the contamination stress-test shows a 47.7% reduction in subspace angle (0.1504 vs. 0.2875). The World Bank 2022 application demonstrates that MacroPCA produces domain-interpretable diagnostics — separating size-driven aggregates from governance anomalies among sovereign states — and reveals a theoretically important nuance: when a dataset is structurally dominated by rowwise outliers that inflate the variance in the clean data, the subspace angle metric can favour the non-robust method while all outlier-detection metrics continue to favour MacroPCA. This underscores the importance of evaluating robust PCA methods on multiple criteria rather than on subspace angle alone.

References

[1] M. Hubert, P. J. Rousseeuw, and W. Van den Bossche, "MacroPCA: An All-in-One PCA Method Allowing for Missing Values as Well as Cellwise and Rowwise Outliers," *Technometrics*, vol. 61, no. 4, pp. 459–473, 2019.

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[4] World Bank, *World Development Indicators*, Washington, DC: The World Bank, 2022. https://databank.worldbank.org/source/world-development-indicators.